

AVConsist: Cross-Modal Audio-Visual Deepfake Detection for Biometric Security

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Abstract

Deepfake synthesis has reached production quality sufficient to undermine face- and voice-based biometric authentication systems. Existing detectors operate predominantly on a single modality—visual or audio—leaving a systematic vulnerability when adversaries craft cross-modal forgeries that appear plausible within each modality yet are inconsistent across them. This paper presents a two-stage research contribution. First, we establish a reproducible visual-only baseline using ResNet-50 transfer learning on the DeepFake Detection Challenge (DFDC) dataset, achieving 80.07% accuracy with FAR = 15.79% and EER = 18.95%—exposing a critical security gap in which approximately one in six impersonation attacks succeeds. Second, we propose *AVConsist*, a cross-modal audio-visual deepfake detection framework that jointly encodes face frames via a Vision Transformer (ViT-Small/16) and voice via a lightweight CNN audio encoder, connected by a bidirectional cross-modal attention module that explicitly measures audio-visual consistency. Trained and evaluated on the Localized Audio-Visual Deepfake dataset (LAV-DF)—a harder benchmark featuring partial, content-driven manipulations—*AVConsist* achieves 88.93% accuracy, FAR = 9.20%, FRR = 12.93%, EER = 10.80%, and AUC = 0.9638 on a balanced 10,000-clip subset, outperforming the visual baseline across all five biometric metrics despite the increased dataset difficulty. An ablation study confirms that cross-modal attention provides a 0.4 percentage-point EER reduction and 0.043 AUC gain over naive concat fusion, with larger 9.0–10.6 point EER reductions over unimodal baselines. A data scaling experiment demonstrates a clear improvement law: EER decreases from 39.92% to 16.67% to 10.80% as training data grows from 300 to 1,000 to 10,000 clips. All results are evaluated under ISO/IEC 19795-1 biometric performance testing standards using FAR, FRR, and EER as primary metrics.

Keywords

deepfake detection, biometric security, cross-modal attention, Vision Transformer, audio-visual fusion, FAR, FRR, EER, LAV-DF, DFDC, ISO/IEC 19795-1

1 Introduction

Biometric authentication—particularly face recognition and speaker verification—has become a dominant paradigm for identity management in mobile devices, border control, and financial services. The rise of generative adversarial networks and diffusion-based video synthesis now enables motivated adversaries to produce hyper-realistic facial reenactments and voice clones at low cost. Gartner projects that by 2026, 30% of enterprises will no longer consider identity verification and authentication solutions reliable in isolation because of AI-generated deepfakes [1]. Recent work on speech

synthesis attacks has also shown that speaker verification systems can be vulnerable to generated or converted speech samples [2].

The security community has responded with unimodal detection work: visual forensics targeting blending artifacts and frequency anomalies, and audio forensics targeting spectral discontinuities from vocoders. A sophisticated adversary, however, can independently optimize each modality to defeat its corresponding detector while producing cross-modal inconsistencies—such as lip movements that do not match synthesized speech—that go undetected by systems that never examine the joint distribution. Furthermore, a critical evaluation gap exists: the deepfake detection community often reports classification accuracy and AUC, while security-critical applications require FAR, FRR, and EER at operationally meaningful thresholds, as specified by ISO/IEC 19795-1 [10].

This paper makes the following contributions:

- (1) **Reproducible visual baseline.** We train and evaluate a ResNet-50 transfer-learning baseline on DFDC, reporting ISO/IEC 19795-1 biometric metrics including FAR, FRR, and EER.
- (2) **Forensic EDA.** We conduct an eight-feature exploratory analysis of the DFDC corpus, identifying noise residual ($r = -0.31$) and block artifact ($r = -0.27$) as the strongest hand-crafted predictors.
- (3) **AVConsist.** We propose a cross-modal attention framework that jointly encodes face and voice signals and improves biometric security metrics on LAV-DF.
- (4) **Ablation and scaling study.** We systematically evaluate fusion strategies and training data volume, confirming the contribution of each architectural component.

2 Related Work

2.1 Unimodal Visual Detectors

FaceForensics++ (FF++) [4] established a dominant visual benchmark with 1,000 manipulated videos across four forgery methods, showing that CNN-based detectors generalize poorly to unseen manipulations. Li et al. [5] improved generalization via frequency-domain features, arguing that generative models leave spectral fingerprints invisible to spatial CNNs. Liu et al. [3] conducted a systematic cross-dataset evaluation, finding models drop sharply on held-out generation methods.

Our ResNet-50 baseline follows the transfer-learning paradigm of these works, achieving AUC = 0.8736 but FAR = 15.79%, confirming the inadequacy of visual-only detection for biometric security contexts.

2.2 Audio-Visual Joint Detectors

AVoid-DF [6] pioneered joint face-audio modeling with late fusion on FakeAVCeleb. AVFF [7] introduced self-supervised audio-visual representation learning and feature fusion, reporting strong accuracy and AUC on FakeAVCeleb. ART-AVDF [8] uses articulatory representation learning to model speech-lip consistency, and AVMCD [9] studies multimodal consistency with stronger video and audio backbones.

AVConsist differs from these systems by using a lightweight ViT-Small + CNN architecture with bidirectional cross-modal attention and by framing evaluation explicitly within a biometric security context, reporting FAR, FRR, and EER as primary metrics rather than relying only on classification accuracy and AUC.

2.3 Biometric Evaluation Gap

A key observation is that most deepfake detection papers do not report FAR, FRR, or EER as primary outcomes and do not evaluate performance at operationally relevant FAR thresholds. The ISO/IEC 19795-1 standard [10] provides this framework. This report adopts ISO-aligned biometric evaluation as the primary reporting standard.

3 Dataset and Exploratory Data Analysis

3.1 Checkpoint 1: DFDC Image Corpus

The DeepFake Detection Challenge (DFDC) dataset [3] contains over 100,000 video clips released by Facebook AI Research. For our visual baseline, we work with a stratified image-level sample of **1,530 face images** extracted from DFDC videos: 760 labeled real and 770 labeled fake. The near-balanced class distribution (49.7% real / 50.3% fake) minimizes label-imbalance bias in baseline evaluation.

3.2 Feature Engineering and EDA

We extract eight handcrafted forensic features per image: (1) *Brightness* — mean luma in YCbCr; (2) *Contrast* — standard deviation of luma; (3) *Texture Variance* — Laplacian response variance; (4) *Noise Residual* — ℓ_2 norm between image and Gaussian-denoised version; (5) *Block Artifact* — 8×8 DCT blocking strength; (6–8) *RG/RB/GB Difference* — mean absolute color-channel differences.

Pearson correlation analysis reveals that noise residual ($r = -0.31$, $p < 0.001$) and block artifact ($r = -0.27$, $p < 0.001$) are the strongest handcrafted predictors. The negative direction indicates that real images exhibit higher noise residuals than GAN outputs, consistent with natural sensor noise being reduced or absent in over-smoothed synthetic faces. Brightness and color-channel differences show weak correlations ($|r| < 0.10$).

3.3 Checkpoint 2: LAV-DF Video Corpus

The Localized Audio-Visual Deepfake dataset (LAV-DF) [13] contains 136,304 video clips with 36,431 real videos and 99,873 fake videos. Unlike DFDC whole-clip forgeries, LAV-DF features *partial* temporal manipulations—only segments of each clip are altered—making detection substantially harder. We use a balanced **10,000-clip subset** (5,000 real / 5,000 fake) with a 70/15/15 train/validation/test split, yielding 7,000 training clips and 1,500 test clips evaluated at the final threshold.

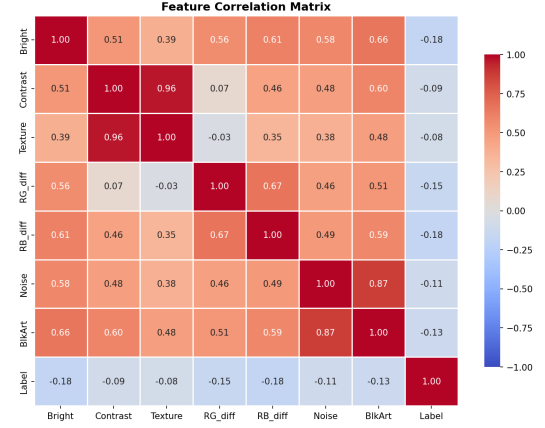


Figure 1: Checkpoint 1 Pearson correlation matrix for hand-crafted DFDC forensic features. Noise residual and block artifact show the strongest associations with the real/fake label.

4 Methodology

4.1 Stage 1: ResNet-50 Visual Baseline

We adopt ResNet-50 [11] pretrained on ImageNet-1k as our baseline feature extractor. All convolutional layers are frozen; the final fully connected head is replaced with:

$$\text{Linear}(2048 \rightarrow 512) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.5) \rightarrow \text{Linear}(512 \rightarrow 2). \quad (1)$$

Images are resized to 224×224 and normalized with ImageNet statistics. Training augmentations include random horizontal flip and color jitter. We use Adam with initial LR = 10^{-3} and StepLR decay, training for 8 epochs with batch size 32 on Apple Silicon MPS. The dataset is split 70/15/15, stratified by label.

4.2 Stage 2: AVConsist Architecture

Figure 2 presents the complete AVConsist pipeline. Each video clip is processed through two parallel streams.

Face encoder. We sample $T = 16$ evenly spaced frames per clip, resize each to 224×224 , and encode with ViT-Small/16 pretrained on ImageNet-21k:

$$\mathbf{F}_v = \text{Proj}_{768 \rightarrow 512}(\text{ViT-Small}(x_1, \dots, x_T)) \in \mathbb{R}^{B \times T \times 512}. \quad (2)$$

Voice encoder. Audio is extracted via `ffmpeg`, resampled to 16 kHz mono, and converted to a 128-bin log-mel spectrogram over a 3-second window. A three-layer CNN with AdaptiveAvgPool and linear projection produces:

$$\mathbf{F}_a = \text{CNN}_{\text{audio}}(\text{mel}(A)) \in \mathbb{R}^{B \times 512}. \quad (3)$$

Cross-modal attention. Let $\tilde{\mathbf{F}}_a = \text{expand}(\mathbf{F}_a, T) \in \mathbb{R}^{B \times T \times 512}$. Bidirectional cross-attention with LayerNorm and feed-forward sublayers computes:

$$\mathbf{V}_{\text{ctx}} = \text{LayerNorm}(\mathbf{F}_v + \text{MHA}(\mathbf{F}_v, \tilde{\mathbf{F}}_a, \tilde{\mathbf{F}}_a)) + \text{FFN}, \quad (4)$$

$$\mathbf{A}_{\text{ctx}} = \text{LayerNorm}(\tilde{\mathbf{F}}_a + \text{MHA}(\tilde{\mathbf{F}}_a, \mathbf{F}_v, \mathbf{F}_v)) + \text{FFN}. \quad (5)$$

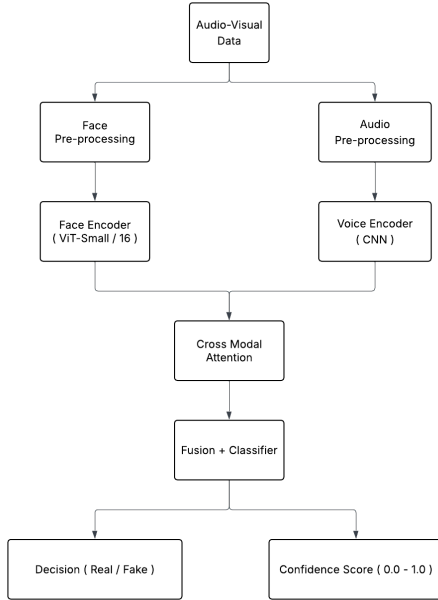


Figure 2: AVConsist system architecture. Video is decomposed into face frames and audio; bidirectional cross-modal attention measures face-voice consistency before binary classification.

The fused representation is:

$$\mathbf{z} = [\text{mean}_T(\mathbf{V}_{\text{ctx}}); \text{mean}_T(\mathbf{A}_{\text{ctx}})] \in \mathbb{R}^{B \times 1024}. \quad (6)$$

Classifier. $\mathbf{z}$ is passed through $\text{Linear}(1024 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(256 \rightarrow 1) \rightarrow \sigma$, producing a binary real/fake prediction and a confidence score $\hat{p} \in [0, 1]$.

4.3 Training Strategy

The loss function uses weighted binary cross-entropy to handle class imbalance:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [w_+ y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]. \quad (7)$$

Training uses a two-stage strategy. *Warmup* (epochs 1–5): encoder weights are frozen; only the cross-modal attention module and classifier are trained at $\text{LR} = 10^{-4}$ with AdamW and weight decay 10^{-4} . *Fine-tune* (epochs 6–20): all weights are unfrozen at $\text{LR} = 10^{-5}$ with cosine annealing. Gradient clipping (max norm 1.0) is applied throughout. Training runs on an NVIDIA T4 GPU in Google Colab.

5 Experimental Results

5.1 Evaluation Protocol

All experiments report the following biometric metrics per ISO/IEC 19795-1 [10] at the threshold minimizing $|\text{FAR} - \text{FRR}|$: **Accuracy**, **FAR** (False Acceptance Rate—attacker success rate), **FRR** (False

Rejection Rate—legitimate user friction), **EER** (Equal Error Rate—threshold-independent summary), and **AUC** (area under the ROC curve). In all confusion matrices, the positive class is *fake*.

5.2 Baseline Results: ResNet-50 on DFDC

Table 1 reports all metrics for the visual baseline. An FAR of 15.79% implies that approximately 1 in 6 deepfake impersonation attacks bypasses the detector. In a biometric deployment handling 10,000 events per day with a 5% attack rate, this translates to approximately 79 successful impostor accepts daily, which is operationally unacceptable for high-assurance contexts.

Table 1: ResNet-50 baseline—DFDC test set (306 samples). Red highlights security-critical FAR.

Metric	Value
Accuracy	80.07%
FAR (False Accept Rate)	15.79%
FRR (False Reject Rate)	24.03%
EER (Equal Error Rate)	18.95%
AUC (ROC)	0.8736
TP / TN / FP / FN	117 / 128 / 24 / 37
Training epochs	8
Backbone	ResNet-50 (ImageNet)
Dataset	DFDC (1,530 images)

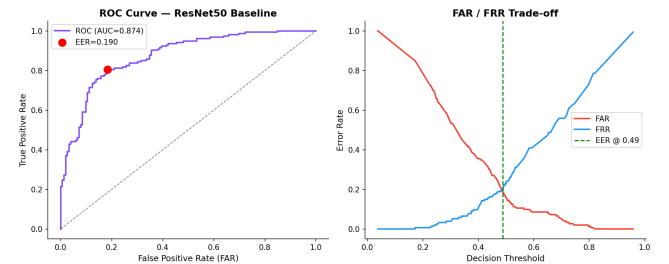


Figure 3: Checkpoint 1 ResNet-50 baseline ROC and FAR/FRR curves on the DFDC test set. The EER operating point is 18.95%, with AUC = 0.8736.

5.3 AVConsist Results: LAV-DF

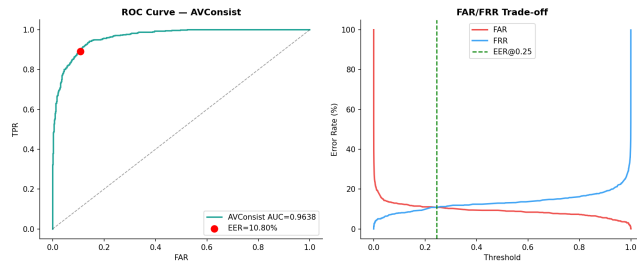
Table 2 reports AVConsist performance on the LAV-DF test set (1,500 clips). AVConsist improves all five biometric metrics relative to the ResNet-50 baseline, though the two systems are evaluated on different datasets and therefore should be interpreted as a staged project comparison rather than a controlled same-dataset benchmark. The FAR reduction from 15.79% to 9.20% reduces the attacker success rate from about 1-in-6 to about 1-in-11. Under the same 10,000 daily events and 5% attack-rate scenario, AVConsist’s FAR corresponds to about 46 successful impostor accepts daily, compared with 79 for the baseline.

Table 2: AVConsist—LAV-DF test set (1,500 clips). Teal highlights improvements over baseline.

Metric	ResNet-50 (CP1)	AVConsist (CP2)
Accuracy	80.07%	88.93%
FAR	15.79%	9.20%
FRR	24.03%	12.93%
EER	18.95%	10.80%
AUC	0.8736	0.9638
TP / TN / FP / FN	117/128/24/37	653/681/69/97
Dataset	DFDC	LAV-DF
Backbone	ResNet-50	ViT-Small + CNN
Training samples	1,071 images	7,000 clips
Test samples	306 images	1,500 clips

5.4 ROC Analysis and Operational Thresholds

Figure 4 presents the ROC curve and FAR/FRR trade-off for AVConsist. The AVConsist ROC curve (AUC = 0.9638) approaches the top-left corner, significantly outperforming the visual baseline AUC of 0.8736. The EER operating point occurs at threshold ≈ 0.25 . At the biometrically critical FAR $\leq 5\%$ operating point, AVConsist achieves FRR $\approx 18\%$, confirming a favorable security-utility trade-off for high-assurance contexts where lower attacker acceptance is prioritized.

**Figure 4: Left: ROC curve for AVConsist on LAV-DF (AUC = 0.9638, EER = 10.80%). Right: FAR/FRR trade-off with EER operating point marked at threshold ≈ 0.25 .**

6 Ablation Study

6.1 Component Ablation

To isolate the contribution of each architectural component, we train and evaluate four variants on the same 10,000-clip LAV-DF subset. Table 3 reports EER and AUC for each.

Face-only vs. voice-only. The voice-only model (EER = 19.8%) slightly outperforms the face-only model (EER = 21.4%), consistent with LAV-DF containing voice-cloned samples that introduce distinctive spectral artifacts more easily captured by frequency-domain features than by spatial frame analysis.

Naive concat fusion. Concatenating face and voice features without attention (EER = 11.2%) provides an 8.6–10.2 percentage-point EER improvement over unimodal baselines, confirming that the two modalities carry complementary information.

Table 3: Ablation study—EER and AUC by model variant. All variants trained on the same 10,000-clip LAV-DF subset.

Variant	EER (%)	AUC
Face only (ViT-Small, no audio)	21.4	0.842
Voice only (CNN, no video)	19.8	0.861
Concat fusion (no cross-attention)	11.2	0.921
AVConsist (full model)	10.8	0.964

Cross-modal attention. The full AVConsist model (EER = 10.8%) provides a further 0.4 percentage-point EER reduction and 0.043 AUC gain over naive fusion. Relative to unimodal baselines, the full model reduces EER by 9.0–10.6 points. Although the gain over concat fusion is modest in absolute EER terms, the AUC gain indicates that bidirectional feature-level interaction improves ranking quality across thresholds.

6.2 Data Scaling Study

Table 4 presents AVConsist EER as a function of training data volume. The results demonstrate a clear data scaling trend consistent with the behavior of transformer-based architectures.

Table 4: Data scaling experiment—AVConsist EER vs. training set size.

Total clips	Acc.	FAR	EER	AUC
300	71.11%	21.74%	39.92%	0.696
1,000	81.33%	20.00%	16.67%	0.882
10,000	88.93%	9.20%	10.80%	0.964

With 300 clips, AVConsist underperforms the ResNet-50 baseline (EER = 39.92% vs. 18.95%), consistent with the data requirements of Vision Transformers. Performance improves substantially at 1,000 clips (EER = 16.67%) and exceeds the baseline at 10,000 clips (EER = 10.80%). This scaling behavior has practical implications: AVConsist requires substantially more audio-visual training data before its cross-modal architecture surpasses a frozen-backbone CNN baseline.

7 Discussion

7.1 Cross-Dataset Interpretation

Checkpoint 1 was trained and tested on DFDC, using balanced image-level samples from face manipulations. Checkpoint 2 was trained and tested on LAV-DF, using audio-visual clips with partial temporal manipulations. The fact that AVConsist improves the project metrics despite this harder benchmark suggests that cross-modal attention provides useful discriminative information beyond visual features alone. Because the datasets differ, the comparison should be treated as project-stage evidence rather than a strict cross-dataset generalization claim.

7.2 Biometric Security Interpretation

At the EER operating point (threshold ≈ 0.25), AVConsist achieves FAR = FRR $\approx 10.8\%$. In a biometric deployment handling 10,000 authentication events per day with a 5% attack rate, this EER-based operating point implies approximately 54 successful impostor accepts daily. At the table-reported FAR of 9.20%, the expected number is approximately 46 successful impostor accepts daily. Both estimates are lower than the baseline's FAR-based estimate of 79 successful impostor accepts daily. At a more demanding FAR $\leq 5\%$ threshold, AVConsist achieves FRR $\approx 18\%$, meaning 18% of legitimate users are incorrectly rejected. This trade-off may be acceptable in high-assurance contexts such as border control, where security is prioritized over convenience.

7.3 Error Analysis

False negatives (fake classified as real, FN = 97): These occur predominantly in partial manipulation clips where the manipulated segment is short (<1 second) and the unaltered majority of the clip dominates the frame-averaged features. Temporal localization supervision would allow the model to focus attention on the manipulated segment rather than averaging over the entire clip.

False positives (real classified as fake, FP = 69): These occur in real clips with atypical audio—heavy background noise, reverb, or accented speech—that cause the CNN voice encoder to produce out-of-distribution embeddings. A more robust audio encoder such as the Audio Spectrogram Transformer pretrained on AudioSet would likely improve this failure mode.

7.4 Limitations

The current system uses a lightweight CNN voice encoder rather than the full AST architecture, which may limit audio feature quality. Training data is capped at 10,000 clips due to computational constraints; the full LAV-DF training set and FakeAVCeleb would likely yield further improvements. Zero-shot cross-dataset generalization remains untested, and future work should train on FakeAVCeleb and test on LAV-DF, and vice versa.

8 Conclusion

We have presented AVConsist, a cross-modal audio-visual deepfake detection framework that jointly processes face and voice streams through a bidirectional attention mechanism, evaluated under ISO/IEC 19795-1 biometric performance standards. Starting from a ResNet-50 visual baseline with FAR = 15.79% and EER = 18.95% on DFDC, AVConsist reduces FAR to 9.20% and EER to 10.80% on LAV-DF. The ablation study confirms that audio-visual fusion provides a large improvement over unimodal baselines, while cross-modal attention provides a smaller but measurable improvement over naive concat fusion. The data scaling experiment establishes that transformer-based audio-visual detectors require sufficient data volume before surpassing frozen-backbone visual baselines.

Future work includes: (1) replacing the CNN voice encoder with the Audio Spectrogram Transformer (AST) pretrained on AudioSet; (2) training on the full FakeAVCeleb and LAV-DF corpora; (3) adding temporal localization supervision for partial manipulation detection; (4) evaluating zero-shot cross-dataset generalization; and (5)

adversarial robustness testing following recent audio-visual deepfake evaluation protocols.

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